CareerLens AI — Smart Career Coach

Task 2 — AI Career Coaching Web Application

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1. Introduction

This project is CareerLens AI, an AI-powered career coaching web application built as part of Task 2 of the Alfido Tech Internship. Rather than a single-purpose tool, CareerLens AI is a full career-assistance platform that combines resume analysis, cover letter generation, interview preparation, skill gap tracking, career roadmapping, and internship fit checking into one dashboard.

The platform is powered by IBM watsonx.ai's Granite large language model, which drives an in-app AI coach chat as well as every analysis feature — from ATS resume scoring to personalized career roadmaps. The live application can be accessed at:

https://web-production-6a332.up.railway.app/

2. Objectives

- Develop a multi-feature AI career-coaching platform rather than a single standalone tool.
- Integrate IBM watsonx.ai's Granite model to power resume analysis, content generation, and conversational coaching.
- Build an ATS resume analyzer that scores a resume against a job description.
- Provide AI-generated cover letters tailored to a specific job description.
- Generate predicted interview questions with sample answers and prep guidance.
- Identify skill gaps against a target role and suggest free learning resources.
- Produce a personalized, step-by-step career roadmap toward a target role.
- Offer an internship fit analyzer for fresher-friendly opportunities.
- Deploy the project online for public access using Railway.

3. Technologies Used

- HTML5
- CSS3
- JavaScript
- IBM watsonx.ai — Granite large language model (AI analysis & generation engine)
- REST API integration for resume, job description, and chat processing
- Railway (deployment and hosting)

4. Website Structure & Features

The homepage presents a dashboard with an embedded AI coach chat panel and quick-action stats (ATS Score, Analyses Done, Avg ATS Score, Saved Jobs), alongside a top navigation menu organizing the platform into six sections:

Dashboard

- Resume upload (PDF, DOCX, or TXT, up to 10MB).
- Live AI Coach chat panel powered by IBM watsonx.ai & Granite for resume tips, interview prep, skill advice, and cover letters.
- Quick-access cards for 'Analyze My Resume' and 'Career Roadmap'.

ATS Analyzer

- Paste a job title, company, and job description to get a match score out of 100.
- Keyword breakdown: matched skills vs. missing keywords.
- Improvement actions and AI-suggested resume bullet rewrites.
- Option to save the job and jump straight to cover letter generation.

Cover Letter

- Generates a tailored, professional cover letter from the uploaded resume and a job description.
- Configurable with the applicant's name.
- Copy and download options for the generated letter.

Interview Prep

- Generates predicted interview questions from a job description.
- Categorizes questions as Behavioral, Technical, and Situational.
- Includes a prep checklist alongside sample answers.

Skill Gaps

- Compares the resume against a target job to surface missing skills.
- Highlights 'Quick Wins' and 'Critical Skill Gaps'.
- Produces a week-by-week learning plan with free learning resources and personalized advice.

Roadmap

- Generates a personalized, step-by-step plan toward a target role.
- Uses the uploaded resume for a tailored roadmap, or produces a general roadmap without one.

Internships

- Internship Fit Analyzer scores how well a fresher's profile matches an internship description.
- Breaks down skills already held versus skills to learn.
- Recommends platforms to apply on.

General Features

- Responsive dashboard-style layout with persistent stats and recent analyses.
- Consistent AI-coach chat experience available across the platform.
- All analysis features powered by the same IBM watsonx.ai Granite backend.

5. Implementation

The front end is built with HTML, CSS, and JavaScript around a dashboard layout, with a resume upload panel feeding into every downstream feature (ATS analysis, cover letters, interview prep, skill gaps, roadmap, internship fit). Each feature submits the resume text and a job description to a REST API, which forwards the content to IBM watsonx.ai's Granite model with a feature-specific prompt.

The model's structured response — score, keyword lists, generated text, or a step-by-step plan — is parsed and rendered back into the relevant panel. The embedded AI Coach chat uses the same Granite backend for open-ended career questions. The application is deployed and hosted on Railway for public access.

6. Project Links

Hosted Link (Live Demo)

https://web-production-6a332.up.railway.app/

GitHub Repository

https://github.com/Meghana544git/CareerLens-AI.git

7. Snippets

Frontend — ATS analysis request (dashboard.js)

```
async function analyzeMatch(resumeFile, jobDescription, jobTitle, company) {
  const formData = new FormData();
formData.append("resume", resumeFile);
  formData.append("jobDescription", jobDescription);
  formData.append("jobTitle", jobTitle);
formData.append("company", company);

  setLoading(true);
  try {
    const res = await fetch("/api/ats-analyze", {
      method: "POST",
body: formData,
    });
    if (!res.ok) throw new Error(`ATS analysis failed: ${res.status}`);

    const result = await res.json();
    renderAtsScore(result);
  } catch (err) {
    showError("Could not analyze resume. Please try again.");
  } finally {
    setLoading(false);
  }
}
```

Backend — REST endpoint calling watsonx.ai Granite (server.js)

```
app.post("/api/ats-analyze", upload.single("resume"), async (req, res) => {
  const resumeText = await extractTextFromFile(req.file);
const { jobDescription, jobTitle, company } = req.body;

  const prompt = buildAtsPrompt({ resumeText, jobDescription, jobTitle, company });

  const watsonRes = await fetch(
    `${process.env.WATSONX_URL}/ml/v1/text/generation?version=2024-05-01`,
    {
      method: "POST",
headers: {
        "Content-Type": "application/json",
        Authorization: `Bearer ${await getAccessToken()}`,
      },
      body: JSON.stringify({
        model_id: "ibm/granite-13b-instruct-v2",
project_id: process.env.WATSONX_PROJECT_ID,
        input: prompt,
        parameters: { max_new_tokens: 600, temperature: 0.2 },
      }),
    }
  );

  const data = await watsonRes.json();
  const parsed = parseAtsResponse(data.results[0].generated_text);

  res.json(parsed); });
```

Parsing the structured AI response (utils.js)

```
function parseAtsResponse(rawText) {
const json = extractJsonBlock(rawText);
   return
{
    score: json.score,                       // 0-100
    matchedSkills: json.matched_skills || [],
missingKeywords: json.missing_keywords || [],
improvementActions: json.improvement_actions || [],
    bulletRewrites: json.bullet_rewrites || [],
  };
}
```

8. Commands

Standard commands for setting up, running, and deploying this type of Node.js + watsonx.ai project on Railway:

Local setup

```
# Clone the repository git
clone <your-repository-url>
cd careerlens-ai

# Install dependencies
npm install

# Configure environment variables
cp .env.example .env
# then set WATSONX_API_KEY, WATSONX_PROJECT_ID, WATSONX_URL

# Run the app locally
npm start
```

Deploy to Railway

```
# Install the Railway CLI (one-time)
npm install -g @railway/cli

# From the project root
railway login
railway init
railway up

# Set environment variables on Railway
railway variables set WATSONX_API_KEY=your_key_here
railway variables set WATSONX_PROJECT_ID=your_project_id
```

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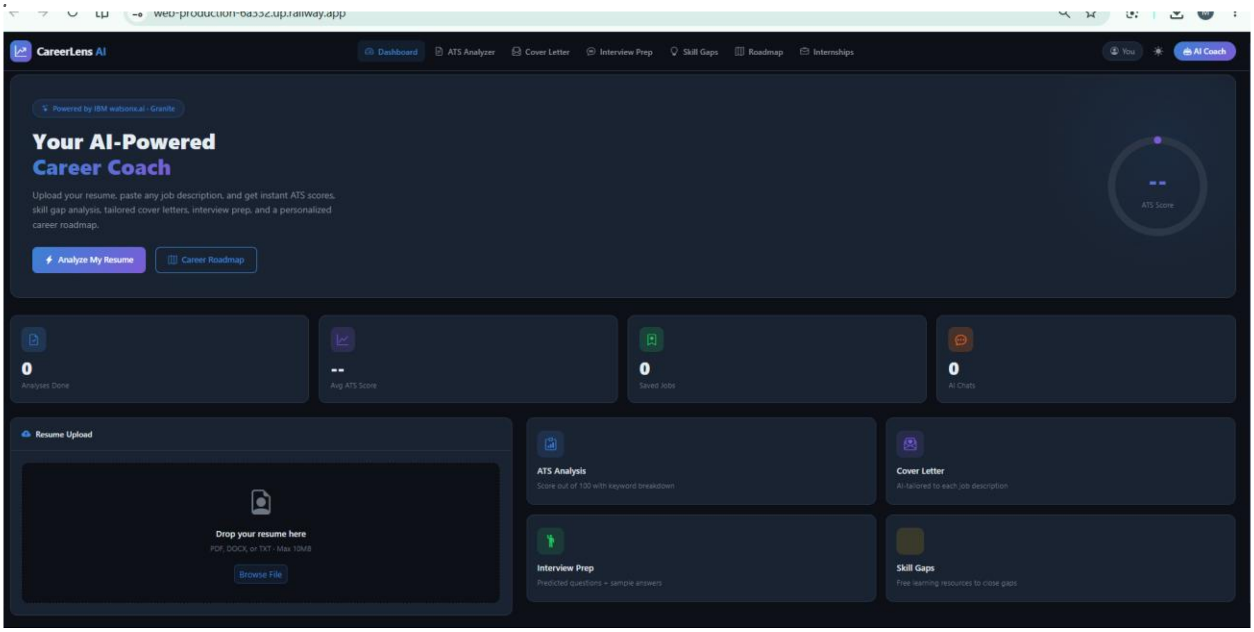
9. Screenshots



10. Learning Outcomes

Through this project, practical experience was gained in:

- Integrating a generative AI model (IBM Granite via watsonx.ai) across multiple distinct features in one platform.
- Designing feature-specific prompts and handling structured AI responses (scores, keyword lists, generated text, roadmaps).
- Handling file uploads (resume parsing) alongside asynchronous API calls and loading states.
- Building a persistent, dashboard-style interface with live stats and recent activity.
- Designing a responsive, multi-feature front end for an AI-driven career tool.
- Deploying a full web application using Railway.

11. Conclusion

CareerLens AI demonstrates the practical application of generative AI across the full job-search workflow — resume analysis, cover letters, interview prep, skill gaps, roadmapping, and internship fit. By integrating IBM watsonx.ai's Granite model through a REST API, the application delivers instant, explainable, and actionable career guidance in one place.

This project strengthened understanding of AI/API integration, multi-feature product design, structured response handling, and deployment of AI-powered web applications for public use.